Rigorous Derivation of Schottky Diode Current

Coupled Poisson–Schrödinger–Landauer Framework

Expanded technical revision with explicit sign conventions, energy bookkeeping, transmission limits, and numerical validation

Purpose. This document expands a compact derivation into a stricter working model suitable for analytical checking and numerical implementation. The core sequence is unchanged: depletion electrostatics → image-force correction → Schrödinger transmission → thermionic state counting → Landauer transport. Added material is explicitly technical: sign conventions, dimensional checks, turning points, the relation between Richardson and Landauer forms, a self-consistent solution loop, and verification tests.

Scope. One-dimensional normal transport through an n-type Schottky contact, with a parabolic effective-mass band as the baseline model. The closed forms are benchmarks, not universal material models.

Contents

1 Notation, coordinate system, and non-negotiable conventions **1**
1.1 Signed field versus field magnitude . . . 1
1.2 Barrier reference . . . 1

2 Depletion electrostatics from Poisson's equation **2**
2.1 Charge model and boundary conditions . . . 2
2.2 Internal checks . . . 2
2.3 Bias dependence: what V_d is and is not . . . 3

3 Barrier construction and image-force lowering **3**
3.1 Local triangular approximation . . . 3
3.2 Using the actual depletion field . . . 3
3.3 The image singularity must be regularized . . . 4

4 Schrödinger transmission **4**
4.1 One-dimensional effective-mass equation . . . 4
4.2 Exact Airy structure for a triangular barrier . . . 4
4.3 WKB transmission for a smooth forbidden region . . . 5
4.4 Turning points with image rounding . . . 5

5 Richardson thermionic current from state counting **6**
5.1 Start from flux, not from a memorized prefactor . . . 6
5.2 Image-lowered thermionic approximation . . . 6

6 Landauer current and its relation to Richardson emission **7**
6.1 Total-energy mode form . . . 7
6.2 Mode-resolved form for a planar Schottky barrier . . . 7
6.3 Ideal diode limit . . . 7

7 Thermionic emission, field emission, and the crossover **7**

8 Self-consistent Poisson–Schrödinger–Landauer solution **8**
8.1 Why a one-pass calculation is not always enough . . . 8
8.2 Recommended nonlinear iteration . . . 8

9 Numerical implementation requirements **9**
9.1 Spatial resolution . . . 9
9.2 Energy integration . . . 9
9.3 Stable scattering calculation . . . 9
9.4 Log-domain WKB . . . 9

10 Illustrative numerical check **9**

11 Verification hierarchy **10**

12 Common failure modes **11**

13 Model limitations and extensions **11**

14 Compact derivation map **12**

References **12**

1 Notation, coordinate system, and non-negotiable conventions

Take the metal/semiconductor interface at $x = 0$ and the semiconductor to occupy $x > 0$. The depletion region extends from 0 to W. Let $q > 0$ be the elementary charge; the electron charge is $-q$.

Symbol	Definition	SI unit
A^{**}	Effective Richardson constant for the chosen band model	$\mathrm{A\,m^{-2}\,K^{-2}}$
$E, E_x, E_\perp$	Total, longitudinal, and transverse electron energies	J
E_C, E_F, μ	Conduction-band edge, Fermi level, electrochemical potential	J
$\mathcal{E}(x)$	Signed electric field, $\mathcal{E} = -d\psi/dx$	$\mathrm{V\,m^{-1}}$
$F(x)$	Field magnitude, $F \equiv \lvert\mathcal{E}\rvert$	$\mathrm{V\,m^{-1}}$
F_s	Surface-field magnitude, $F(0)$	$\mathrm{V\,m^{-1}}$
f_S, f_M	Semiconductor and metal Fermi occupations	–
$h, \hbar$	Planck constants, $\hbar = h/(2\pi)$	J s
J, J_S	Current density and saturation-current density	$\mathrm{A\,m^{-2}}$
k_B, T	Boltzmann constant and absolute temperature	$\mathrm{J\,K^{-1}}$, K
m^*, m_0	Effective and free-electron masses	kg
$M(E)/A$	Propagating modes per unit area at total energy E	$\mathrm{m^{-2}}$
N_D	Ionized donor concentration in the depletion approximation	$\mathrm{m^{-3}}$
$U(x)$	Electron potential-energy barrier	J
V, V_d	Applied voltage and semiconductor depletion-potential drop	V
W	Depletion width	m
ε_s	Semiconductor permittivity	$\mathrm{F\,m^{-1}}$
$\Phi_\mathrm{B}, \Delta\Phi_\mathrm{B}$	Barrier height and image-force lowering, both expressed in volts	V
$\mathcal{T}$	Transmission probability	–
ψ, Ψ	Electrostatic potential and quantum wavefunction	V, –

Strict modeling rule

Never mix a barrier in volts with an energy in joules. Φ_B and $\Delta\Phi_\mathrm{B}$ are voltages; the associated electron energies are $q\Phi_\mathrm{B}$ and $q\Delta\Phi_\mathrm{B}$. Every exponential must therefore contain an energy divided by $k_\mathrm{B}T$, e.g. $\exp[-q\Phi_\mathrm{B}/(k_\mathrm{B}T)]$.

1.1 Signed field versus field magnitude

With x increasing into the semiconductor and positive depletion charge $+qN_D$, Poisson's equation gives a signed field that points toward the metal and is therefore negative. It is useful to keep the sign and magnitude separate:

$$\mathcal{E}(x) = -\frac{d\psi}{dx}, \qquad F(x) = |\mathcal{E}(x)|. \tag{1}$$

This avoids a common inconsistency in compact derivations, where $F = -d\psi/dx$ is declared but a positive formula for $F(x)$ is then used as though it were the signed field.

1.2 Barrier reference

Define the nominal Schottky barrier by

$$q\Phi_\mathrm{B} \equiv E_C(0) - E_F, \tag{2}$$

unless a different reference is explicitly stated. This definition already contains the bulk separation $E_C - E_F$ plus the electrostatic band bending up to the interface.

Do not skip this

If $q\Phi_{\mathrm{B}} = E_C(0) - E_F$ is used, do not multiply the final thermionic current by a second factor $\exp[-(E_C - E_F)/(k_{\mathrm{B}}T)]$. That would count the Fermi-to-conduction-band separation twice.

2 Depletion electrostatics from Poisson's equation

2.1 Charge model and boundary conditions

For a uniformly doped, fully depleted n-type semiconductor,

$$\rho(x) = \begin{cases} +qN_D, & 0 < x < W, \\ 0, & x \geq W. \end{cases} \tag{3}$$

Assume constant ε_{s}. Poisson's equation is

$$\frac{d^2\psi}{dx^2} = -\frac{\rho}{\varepsilon_{\mathrm{s}}} = -\frac{qN_D}{\varepsilon_{\mathrm{s}}}. \tag{4}$$

The neutral edge satisfies $\mathcal{E}(W) = 0$. Integrating equation (4),

$$\mathcal{E}(x) = -\frac{d\psi}{dx} = \frac{qN_D}{\varepsilon_{\mathrm{s}}}(x - W) \leq 0, \tag{5}$$

$$F(x) = |\mathcal{E}(x)| = \frac{qN_D}{\varepsilon_{\mathrm{s}}}(W - x), \tag{6}$$

$$F_s = F(0) = \frac{qN_D W}{\varepsilon_{\mathrm{s}}}. \tag{7}$$

The depletion potential is the electrostatic rise from interface to neutral bulk,

$$V_d \equiv \psi(W) - \psi(0) = -\int_0^W \mathcal{E}(x)\,dx = \int_0^W F(x)\,dx \tag{8}$$

$$= \frac{qN_D W^2}{2\varepsilon_{\mathrm{s}}}. \tag{9}$$

Hence

$$\boxed{W = \sqrt{\frac{2\varepsilon_{\mathrm{s}} V_d}{qN_D}}}, \qquad \boxed{F_s = \sqrt{\frac{2qN_D V_d}{\varepsilon_{\mathrm{s}}}}}. \tag{10}$$

2.2 Internal checks

The integrated ionized donor charge per unit area is

$$Q_D = qN_D W, \tag{11}$$

so Gauss's law gives $F_s = Q_D/\varepsilon_{\mathrm{s}}$, consistent with equation (7). Also,

$$\frac{1}{2}F_s W = \frac{qN_D W^2}{2\varepsilon_{\mathrm{s}}} = V_d, \tag{12}$$

which is simply the area under the linearly varying field profile.

Dimensional check

$\varepsilon_{\mathrm{s}} V_d/(qN_D)$ has units m^2, so W has units of length. Likewise $qN_D V_d/\varepsilon_{\mathrm{s}}$ has units $(\mathrm{V/m})^2$, so F_s is a field.

2.3 Bias dependence: what V_d is and is not

For a simple forward-biased n-type Schottky contact one often writes, approximately,

$$V_d(V_s) \approx V_d(0) - V_s, \tag{13}$$

where V_s is the portion of the applied bias that actually drops across the semiconductor depletion region. Terminal voltage need not equal V_s when series resistance, interface layers, or other voltage drops are present.

Strict modeling rule

Do not replace V_d by the terminal voltage without an electrostatic argument. The Poisson solution only knows the potential drop across the region being solved.

3 Barrier construction and image-force lowering

3.1 Local triangular approximation

Close enough to the interface that the field varies little, replace $F(x)$ by F_s. The electron barrier energy is then approximated by

$$U(x) = q\Phi_{\mathrm{B}} - qF_s x - \frac{q^2}{16\pi\varepsilon_{\mathrm{s}} x}, \qquad x > 0. \tag{14}$$

The final term is the classical image potential for a planar metal interface in a continuum dielectric.

The barrier maximum satisfies $dU/dx = 0$:

$$-qF_s + \frac{q^2}{16\pi\varepsilon_{\mathrm{s}} x^2} = 0, \tag{15}$$

so

$$\boxed{x_m = \sqrt{\frac{q}{16\pi\varepsilon_{\mathrm{s}} F_s}}}. \tag{16}$$

At $x = x_m$, the field term and image term have equal magnitude. Therefore

$$U(x_m) = q\Phi_{\mathrm{B}} - 2qF_s x_m = q(\Phi_{\mathrm{B}} - \Delta\Phi_{\mathrm{B}}), \tag{17}$$

$$\boxed{\Delta\Phi_{\mathrm{B}}} = 2F_s x_m = \boxed{\sqrt{\frac{qF_s}{4\pi\varepsilon_{\mathrm{s}}}}}, \tag{18}$$

$$\boxed{\Phi_{\mathrm{B},eff}} = \Phi_{\mathrm{B}} - \Delta\Phi_{\mathrm{B}}. \tag{19}$$

The energy lowering is $q\Delta\Phi_{\mathrm{B}} = \sqrt{q^3 F_s/(4\pi\varepsilon_{\mathrm{s}})}$.

3.2 Using the actual depletion field

The local triangular barrier is a near-interface approximation. The depletion-field magnitude is

$$F(x) = F_s - \frac{qN_D}{\varepsilon_{\mathrm{s}}} x, \tag{20}$$

so the electrostatic reduction of the electron barrier from 0 to x is

$$q\int_0^x F(s)\,ds = qF_s x - \frac{q^2 N_D}{2\varepsilon_{\mathrm{s}}} x^2. \tag{21}$$

A more faithful depletion-plus-image barrier is therefore

$$U_{\mathrm{dep+im}}(x) = q\Phi_{\mathrm{B}} - qF_s x + \frac{q^2 N_D}{2\varepsilon_{\mathrm{s}}} x^2 - \frac{q^2}{16\pi\varepsilon_{\mathrm{s}} x}. \tag{22}$$

Near the surface, the quadratic term is small compared with $qF_s x$ when $x \ll W$, which recovers equation (14).

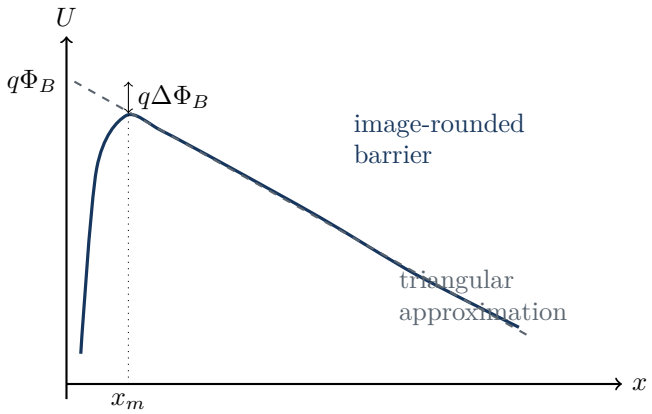


Figure 1: Schematic barrier. The image term rounds and lowers the maximum relative to the purely triangular electrostatic barrier. The $x \to 0$ divergence is not physically valid at atomic distances.

3.3 The image singularity must be regularized

The continuum expression $-q^2/(16\pi\varepsilon_s x)$ diverges as $x \to 0$. It cannot be extrapolated to the atomic interface. A numerical model must state how this region is treated. Common consistent choices include:

- an image-plane offset $x \mapsto x + x_0$ with a documented x_0;
- a short-distance cutoff matched to an interface potential;
- an atomistic or first-principles interface model replacing the continuum image term.

Do not skip this

A result that depends strongly on an undocumented image cutoff is not a controlled prediction. Report the cutoff or interface model and perform a sensitivity test.

4 Schrödinger transmission

4.1 One-dimensional effective-mass equation

For normal transport with constant effective mass m^*,

$$\left[-\frac{\hbar^2}{2m^*}\frac{d^2}{dx^2} + U(x)\right]\Psi(x) = E_x\Psi(x). \tag{23}$$

Here E_x is the longitudinal energy associated with motion normal to the interface. When transverse momentum is conserved at a planar interface, barrier crossing is controlled by E_x, not by the total kinetic energy alone.

If m^* changes with position, a safer envelope-function operator is the BenDaniel–Duke form

$$-\frac{\hbar^2}{2}\frac{d}{dx}\left[\frac{1}{m^*(x)}\frac{d\Psi}{dx}\right] + U(x)\Psi = E_x\Psi, \tag{24}$$

with the corresponding interface matching conditions. The constant-mass equation equation (23) is the baseline closed-form model.

4.2 Exact Airy structure for a triangular barrier

Let

$$U(x) = U_0 - qFx, \tag{25}$$

with constant $F > 0$. The classical turning point is

$$x_t = \frac{U_0 - E_x}{qF}. \tag{26}$$

Define the Airy length

$$\ell = \left(\frac{\hbar^2}{2m^* qF}\right)^{1/3} \tag{27}$$

and dimensionless coordinate

$$z = \frac{x_t - x}{\ell}. \tag{28}$$

Then equation (23) becomes

$$\frac{d^2\Psi}{dz^2} - z\Psi = 0, \tag{29}$$

whose independent solutions are $\mathrm{Ai}(z)$ and $\mathrm{Bi}(z)$. Exact transmission follows by matching wavefunctions and fluxes to the adjoining regions.

4.3 WKB transmission for a smooth forbidden region

Between turning points x_1 and x_2 satisfying $U(x_i) = E_x$,

$$\kappa(x) = \frac{\sqrt{2m^*[U(x) - E_x]}}{\hbar}. \tag{30}$$

The leading exponential WKB result is

$$\boxed{\mathcal{T}(E_x) \approx \exp\left[-2\int_{x_1}^{x_2} \kappa(x)\,dx\right]} = \exp\left[-\frac{2}{\hbar}\int_{x_1}^{x_2} \sqrt{2m^*[U(x) - E_x]}\,dx\right]. \tag{31}$$

For the triangular barrier, $x_1 = 0$ and $x_2 = x_t$ in the simple idealization, giving

$$\boxed{\mathcal{T}(E_x) \approx \exp\left[-\frac{4\sqrt{2m^*}(U_0 - E_x)^{3/2}}{3\hbar qF}\right]}, \qquad E_x < U_0. \tag{32}$$

Above the barrier, a pure thermionic model takes $\mathcal{T} \approx 1$; a quantum scattering calculation can retain above-barrier reflection if present.

4.4 Turning points with image rounding

For the constant-field image barrier in equation (14), define

$$a \equiv \frac{q^2}{16\pi\varepsilon_s}. \tag{33}$$

Turning points solve

$$qF_s x^2 - (q\Phi_B - E_x)x + a = 0, \tag{34}$$

so

$$x_{1,2} = \frac{(q\Phi_B - E_x) \mp \sqrt{(q\Phi_B - E_x)^2 - 4qF_s a}}{2qF_s}. \tag{35}$$

Two real turning points exist only when E_x is below the image-lowered maximum $q(\Phi_B - \Delta\Phi_B)$. The WKB action for the image-rounded barrier is generally evaluated numerically using equation (31).

Strict modeling rule

Do not use the triangular Fowler–Nordheim exponent with an image-rounded barrier and then also insert a separate image-lowering factor unless you have shown that the approximation does not double-count the same barrier modification.

5 Richardson thermionic current from state counting

5.1 Start from flux, not from a memorized prefactor

For a 3D isotropic parabolic band with spin degeneracy $g_s = 2$, the one-way particle flux toward the interface is

$$\Gamma_+ = \frac{g_s}{(2\pi)^3} \int_{k_x>0} v_x f(E)\, d^3k, \qquad v_x = \frac{\hbar k_x}{m^*}. \tag{36}$$

To make the barrier reference explicit, write

$$E = E_{C,b} + E_x + E_\perp, \tag{37}$$

where $E_{C,b}$ is the bulk semiconductor conduction-band edge. Under nondegenerate statistics,

$$f(E) \approx \exp\left[-\frac{E - E_F}{k_\mathrm{B}T}\right]. \tag{38}$$

Let the electrostatic band bending from bulk to the interface be

$$\Delta E_C = E_C(0) - E_{C,b}. \tag{39}$$

For a planar barrier that conserves transverse momentum, thermionic escape requires

$$E_x \geq \Delta E_C. \tag{40}$$

Then the flux separates into longitudinal and transverse integrals:

$$\Gamma_+ = \frac{2}{(2\pi)^3} \exp\left[-\frac{E_{C,b} - E_F}{k_\mathrm{B}T}\right] \left(\iint e^{-E_\perp/(k_\mathrm{B}T)} dk_y\, dk_z\right) \left(\int_{k_{x0}}^{\infty} v_x e^{-E_x/(k_\mathrm{B}T)} dk_x\right). \tag{41}$$

The transverse Gaussian integral is

$$\iint e^{-E_\perp/(k_\mathrm{B}T)} dk_y\, dk_z = \frac{2\pi m^* k_\mathrm{B}T}{\hbar^2}, \tag{42}$$

and because $dE_x = \hbar v_x\, dk_x$,

$$\int_{k_{x0}}^{\infty} v_x e^{-E_x/(k_\mathrm{B}T)} dk_x = \frac{k_\mathrm{B}T}{\hbar} \exp\left[-\frac{\Delta E_C}{k_\mathrm{B}T}\right]. \tag{43}$$

Therefore

$$J_S = q\Gamma_+ = \frac{q m^* k_\mathrm{B}^2 T^2}{2\pi^2\hbar^3} \exp\left[-\frac{E_C(0) - E_F}{k_\mathrm{B}T}\right] \tag{44}$$

$$= A^{**} T^2 \exp\left[-\frac{q\Phi_\mathrm{B}}{k_\mathrm{B}T}\right], \tag{45}$$

with

$$\boxed{A^{**} = \frac{4\pi q m^* k_\mathrm{B}^2}{h^3}} = \frac{q m^* k_\mathrm{B}^2}{2\pi^2\hbar^3}. \tag{46}$$

5.2 Image-lowered thermionic approximation

If image-force lowering is treated only as a shift of the barrier maximum, then

$$J_S^{(\mathrm{IF})} \approx A^{**} T^2 \exp\left[-\frac{q(\Phi_\mathrm{B} - \Delta\Phi_\mathrm{B})}{k_\mathrm{B}T}\right] = J_S \exp\left(\frac{q\Delta\Phi_\mathrm{B}}{k_\mathrm{B}T}\right). \tag{47}$$

This is a thermionic approximation. A tunneling calculation through the full rounded barrier should instead place the modified $U(x)$ directly into $\mathcal{T}(E_x)$.

Strict modeling rule

Richardson's law is the high-barrier, nondegenerate, thermionic benchmark of this band model. It is not an independent current channel to be added on top of a Landauer calculation that already uses the same transmitting states.

6 Landauer current and its relation to Richardson emission

6.1 Total-energy mode form

A compact current-density form is

$$\boxed{J = \frac{q}{h}\int M_A(E)\,\overline{\mathcal{T}}(E,V)\,[f_S(E) - f_M(E)]\,dE}, \tag{48}$$

where $M_A(E) \equiv M(E)/A$ is the number of propagating modes per unit area and $\overline{\mathcal{T}}$ is the transmission appropriately averaged over those modes.

For a 3D isotropic parabolic band with spin degeneracy two,

$$\boxed{M_A(E) = \frac{m^*}{\pi\hbar^2}(E - E_C)\,\Theta(E - E_C)}. \tag{49}$$

This expression counts transverse channels whose threshold lies below the total energy E.

6.2 Mode-resolved form for a planar Schottky barrier

Because a planar barrier naturally depends on longitudinal energy, the more transparent expression is

$$J = \frac{q}{h}\frac{g_s}{(2\pi)^2}\int d^2k_\perp \int dE_x\,\mathcal{T}(E_x, k_\perp, V)\,[f_S(E_x, E_\perp) - f_M(E_x, E_\perp)]\,. \tag{50}$$

If the barrier and masses are translationally invariant parallel to the interface, $k_\perp$ is conserved. In the simplest isotropic model, $\mathcal{T}$ depends mainly on E_x.

6.3 Ideal diode limit

Adopt the voltage convention

$$\mu_S - \mu_M = qV, \tag{51}$$

with positive V corresponding to the standard forward-bias sign in the diode equation. For ideal thermionic transmission, nondegenerate injection, and detailed balance, the two opposing one-way fluxes differ by the Boltzmann factor $\exp(qV/k_\mathrm{B}T)$, yielding

$$\boxed{J = J_S\left[\exp\left(\frac{qV}{k_\mathrm{B}T}\right) - 1\right]}, \qquad J_S = A^{**}T^2\exp\left[-\frac{q\Phi_\mathrm{B}}{k_\mathrm{B}T}\right]. \tag{52}$$

This is the ideality-factor $n = 1$ limit.

Do not skip this

A Landauer calculation is only quantitatively meaningful when the mode normalization, reservoir electrochemical potentials, voltage sign, barrier profile, and the energy argument of $\mathcal{T}$ all use the same reference. A formula can look dimensionally correct and still violate detailed balance if those references are mixed.

7 Thermionic emission, field emission, and the crossover

The barrier does not create separate species called "thermionic" and "tunneling" electrons. They are limiting portions of one energy-resolved transport integral:

- **Thermionic emission:** energies at or above the barrier maximum, where $\mathcal{T} \approx 1$ in the ideal limit.

- **Field emission:** electrons well below the barrier maximum that tunnel through a thin barrier.
- **Thermionic-field emission:** the intermediate region where thermal occupation lifts carriers partway up the barrier and tunneling completes the passage.

A unified Landauer integral with the actual $\mathcal{T}(E_x, V)$ handles all three without an arbitrary decomposition.

A useful dimensionless WKB diagnostic is the action

$$S(E_x) = 2\int_{x_1}^{x_2} \kappa(x)\, dx, \qquad \mathcal{T} \sim e^{-S}. \tag{53}$$

When $S \gg 1$, tunneling is exponentially suppressed; when $S \lesssim 1$, the barrier is highly transparent and a leading-order WKB approximation becomes less trustworthy near the turning points.

8 Self-consistent Poisson–Schrödinger–Landauer solution

8.1 Why a one-pass calculation is not always enough

The depletion solution assumes the mobile carrier density is negligible inside the space-charge region. When tunneling, accumulation, inversion, degeneracy, or strong nonequilibrium carrier injection matters, the charge density should be updated rather than fixed at qN_D.

For constant permittivity, a more general Poisson equation is

$$\frac{d^2\psi}{dx^2} = -\frac{\rho(x)}{\varepsilon_s}, \qquad \rho = q\left[N_D^+ - N_A^- + p - n\right] + \rho_{\text{fixed}}. \tag{54}$$

The electron density $n(x)$ should be computed from the same quantum states or Green-function/scattering-state description used for transport, with reservoir occupations applied consistently.

8.2 Recommended nonlinear iteration

A robust implementation can be organized as follows.

1. **Define references and boundaries.** Fix the contact electrochemical potentials, potential zero, conduction-band offsets, device length, and boundary conditions.
2. **Initialize $\psi^{(0)}(x)$.** The depletion solution is a good initial guess when applicable.
3. **Construct $U^{(i)}(x)$.** Build the conduction-band/electron potential-energy profile from $\psi^{(i)}$, then add interface terms such as an explicitly regularized image potential.
4. **Solve quantum transport.** Obtain $\mathcal{T}^{(i)}(E_x)$ and, if performing full charge self-consistency, the occupied scattering states or local spectral density.
5. **Update carrier density.** Compute $n^{(i)}(x)$ from both reservoirs and then $\rho^{(i)}(x)$.
6. **Resolve Poisson.** Solve equation (54) for a raw update $\psi_{\text{raw}}^{(i+1)}$.
7. **Mix.** Use under-relaxation,

$$\psi^{(i+1)} = (1-\alpha)\psi^{(i)} + \alpha\psi_{\text{raw}}^{(i+1)}, \qquad 0 < \alpha \le 1, \tag{55}$$

   with smaller α when the iteration oscillates.
8. **Converge.** Require both potential and charge/current residuals to meet stated tolerances.
9. **Integrate current.** Evaluate the converged Landauer current and repeat for the desired voltage/temperature sweep.

Strict modeling rule

Do not insert the image self-energy into Poisson as though it were a real semiconductor space-charge density. Treat the image term as part of the electron single-particle barrier unless a more complete electrodynamic model explicitly says otherwise.

9 Numerical implementation requirements

9.1 Spatial resolution

The mesh must resolve the shortest relevant length scale. Typical candidates are

$$\ell_F = \left(\frac{\hbar^2}{2m^* qF}\right)^{1/3}, \qquad x_m = \sqrt{\frac{q}{16\pi\varepsilon_s F_s}}, \qquad \lambda_{\mathrm{dB}} \sim \frac{2\pi\hbar}{\sqrt{2m^* E}}. \tag{56}$$

A grid that resolves W but not x_m or ℓ_F can produce an apparently smooth potential while badly miscalculating tunneling.

9.2 Energy integration

The energy grid must cover both occupied reservoir tails and the transmission turn-on. At minimum:

- extend several $k_{\mathrm{B}}T$ beyond the larger reservoir chemical potential;
- refine around the barrier maximum and any sharp resonance or transmission onset;
- check that enlarging the integration window changes J negligibly.

Adaptive quadrature is usually preferable when $\mathcal{T}(E)$ spans many decades.

9.3 Stable scattering calculation

Naive transfer-matrix multiplication can overflow or underflow for thick forbidden regions because growing and decaying exponentials coexist. Prefer a numerically stable scattering-matrix, log-derivative, recursive Green-function, or carefully scaled Airy/WKB formulation.

For a conservative single-channel problem, verify at every energy that

$$R(E) + T(E) = 1 \tag{57}$$

within numerical tolerance after proper velocity normalization.

9.4 Log-domain WKB

When $S(E_x)$ is large, compute

$$\ln \mathcal{T} = -S \tag{58}$$

rather than forming $\exp(-S)$ prematurely. This prevents numerical underflow and makes convergence of exponentially small contributions easier to diagnose.

10 Illustrative numerical check

This example is deliberately simple and uses the isotropic single-mass model only. Take

$$T = 300\,\mathrm{K}, \qquad N_D = 1\times10^{16}\,\mathrm{cm}^{-3}, \qquad \varepsilon_{\mathrm{s}} = 11.7\,\varepsilon_0,$$
$$V_d = 0.70\,\mathrm{V}, \qquad \Phi_{\mathrm{B}} = 0.80\,\mathrm{V}, \qquad m^* = 0.26\,m_0.$$

Then equation (10) gives

$$W \approx 3.01 \times 10^{-7}\,\mathrm{m} = 301\,\mathrm{nm}, \qquad F_s \approx 4.65 \times 10^{6}\,\mathrm{V\,m^{-1}}. \tag{59}$$

Image-force quantities are

$$x_m \approx 2.57\,\mathrm{nm}, \qquad \Delta\Phi_{\mathrm{B}} \approx 2.39 \times 10^{-2}\,\mathrm{V}. \tag{60}$$

Thus $x_m/W \approx 8.5 \times 10^{-3}$, so the locally constant-field approximation is geometrically plausible for this particular example.

The Airy length at F_s is

$$\ell_F = \left(\frac{\hbar^2}{2m^* q F_s}\right)^{1/3} \approx 3.16\,\mathrm{nm}, \tag{61}$$

which is comparable to x_m. A tunneling mesh therefore needs nanometre-scale resolution near the interface even though the depletion width is hundreds of nanometres.

The single-isotropic-mass Richardson constant is

$$A^{**} = \frac{4\pi q m^* k_{\mathrm{B}}^2}{h^3} \approx 31.2\,\mathrm{A\,cm^{-2}\,K^{-2}}. \tag{62}$$

The nominal saturation current density is

$$J_S \approx 1.02 \times 10^{-7}\,\mathrm{A\,cm^{-2}}, \tag{63}$$

and using only the image-lowered thermionic barrier gives

$$J_S^{(\mathrm{IF})} \approx 2.58 \times 10^{-7}\,\mathrm{A\,cm^{-2}}, \tag{64}$$

a factor of about 2.52 larger at 300 K.

Do not skip this

These numbers are not a quantitative silicon contact prediction. Real Si requires appropriate valley degeneracy, anisotropic transport and density-of-states masses, interface properties, and a material/contact-specific barrier. The example is a consistency check for the algebra and numerical scales only.

11 Verification hierarchy

A serious calculation should pass the following tests in order. Failure at an earlier level invalidates confidence in later results.

1. **Units.** Every term in U is an energy; every exponent is dimensionless; J has units $\mathrm{A\,m^{-2}}$.
2. **Electrostatics.** In the depletion limit, recover equation (10); check $F(W) = 0$ and $V_d = \frac{1}{2}F_s W$.
3. **Barrier.** Recover equations (16) and (19); verify that the chosen image regularization does not spuriously dominate the action.
4. **Quantum solver.** For test barriers with known solutions, recover expected Airy/rectangular-barrier transmission and enforce flux conservation.
5. **Equilibrium.** At $V = 0$ with equal temperature and consistent reservoirs, the net Landauer current must vanish to numerical precision.
6. **Thermionic limit.** Suppress tunneling and recover the Richardson law in equation (45).
7. **Ideal bias limit.** Under the stated nondegenerate ideal assumptions, recover equation (52) with $n = 1$.
8. **Grid convergence.** Refine both x and energy meshes until J and relevant barrier metrics stop changing at the claimed precision.
9. **Parameter sensitivity.** Vary the interface cutoff, mass, barrier height, and boundary placement over justified ranges and report which assumptions dominate uncertainty.

12 Common failure modes

Failure	Why it is wrong	Correction
Using F as both signed field and positive magnitude	Produces silent sign contradictions in Poisson integration	Keep $\mathcal{E} = -d\psi/dx$ signed and define $F = \lvert\mathcal{E}\rvert$
Putting Φ_B directly into an energy exponent without q	Mixes volts and joules	Use $q\Phi_B/(k_B T)$
Multiplying by both $e^{-q\Phi_B/k_B T}$ and $e^{-(E_C-E_F)/k_B T}$	Double-counts the same band-to-Fermi separation if $\Phi_B = [E_C(0) - E_F]/q$	Choose one explicit energy reference and derive from it
Using total energy in a transmission that should depend on E_x	Violates transverse-momentum bookkeeping for a planar barrier	Use mode-resolved Landauer transport or a correctly averaged $\overline{\mathcal{T}}(E)$
Adding Richardson current to a Landauer current	Counts the same thermionic states twice	Treat Richardson as a limiting check of Landauer
Using the $1/x$ image potential down to $x = 0$	Applies a continuum model outside its validity and introduces a singularity	State and test a cutoff/interface model
Using terminal V as V_d by default	Ignores series/interface voltage drops	Solve or model the voltage partition
No equilibrium test	Can hide reservoir-reference and sign errors	Require $J(V = 0) = 0$

13 Model limitations and extensions

The compact closed forms rely on assumptions that should be listed, not hidden:

- isotropic parabolic effective mass;
- one-dimensional barrier variation and a planar interface;
- depletion approximation for the analytical Poisson solution;
- ideal or explicitly parameterized interface chemistry;
- continuum image-force model with a stated short-distance treatment;
- coherent single-particle transmission for the Landauer step;
- WKB only where the barrier is sufficiently smooth and the action is not too small;
- nondegenerate statistics for the simple Richardson derivation.

For quantitative Si, 4H-SiC, GaN, or other materials, use material-specific valley structure, anisotropic masses, dielectric response, doping ionization, interface states, barrier inhomogeneity, and experimentally justified barrier data. If scattering inside the barrier or contact region is important, extend the transport treatment beyond a purely coherent one-dimensional transmission model.

14 Compact derivation map

For reference, the entire chain can be summarized without losing the critical definitions:

Poisson:
$$W = \sqrt{\frac{2\varepsilon_s V_d}{qN_D}}, \quad F_s = \sqrt{\frac{2qN_D V_d}{\varepsilon_s}}, \tag{65}$$

Image lowering:
$$\Delta\Phi_B = \sqrt{\frac{qF_s}{4\pi\varepsilon_s}}, \quad \Phi_{B,eff} = \Phi_B - \Delta\Phi_B, \tag{66}$$

WKB:
$$\mathcal{T}(E_x) \approx \exp\left[-\frac{2}{\hbar}\int_{x_1}^{x_2}\sqrt{2m^*[U(x)-E_x]}\,dx\right], \tag{67}$$

Richardson:
$$J_S = A^{**}T^2 e^{-q\Phi_B/(k_B T)}, \quad A^{**} = \frac{4\pi q m^* k_B^2}{h^3}, \tag{68}$$

Landauer:
$$J = \frac{q}{h}\int M_A(E)\overline{\mathcal{T}}(E,V)[f_S - f_M]dE, \tag{69}$$

Ideal limit:
$$J = J_S\left[e^{qV/(k_B T)} - 1\right]. \tag{70}$$

Strict modeling rule

The order matters. Electrostatics defines the barrier; the barrier defines transmission; transmission and reservoir occupations define current. Any shortcut that changes one layer must be propagated consistently through the layers below it.

References

[1] R. T. Tung, "The physics and chemistry of the Schottky barrier height," *Applied Physics Reviews* **1**, 011304 (2014). DOI: 10.1063/1.4858400.

[2] S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed., Wiley (2007).

Expanded from the supplied compact Poisson–Schrödinger–Landauer note. Additional derivational steps, sign-resolution, numerical guidance, and the illustrative calculation are part of this revision.